Resolution Landscapes in WPM

Causal construction, pocket geometry, stability and temporal use

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Abstract

A resolution landscape maps admissible representation choices to estimated target-specific risk. This study derives its construction from issued forecasts and matured outcomes, distinguishes empirical risk from population risk, and develops connected pockets, flow cross-sections, coordinate metrics and temporal lineage. Elementary stability results bound selection error under imperfect landscapes and explain why amplitude accuracy need not preserve basin identity. A local polynomial calculation illustrates the window–noise trade-off without proposing a universal optimal window. Spatial fields and probability maps are treated as separate mathematical objects.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 The forecasting query comes before the landscape

Assume familiarity with conditional expectation, regression and elementary topology on a grid. Let $\mathcal{F}_t$ be the information available at time t. Choose a target functional T, its outcome $Y_{t,P}^{T}$, a horizon $P > 0$, and a loss ℓ_T. A target may be a future value, an event anywhere before $t + P$, or a future spatial field. These require different labels and losses.

A lens $r = (W, m, \lambda)$ chooses past extent W, internal resolution m and optional model setting λ. Its forecast $\widehat{Y}_{s,P,r}$ must be $\mathcal{F}_s$-measurable. This does not require deterministic forecasts, but

any randomization must be generated using information available at s. The ideal conditional landscape is

$$L_t^*(r; T, P) = \mathbb{E}[\ell_T(Y_{t,P}^T, \widehat{Y}_{t,P,r}) \mid \mathcal{F}_t]. \tag{1.1}$$

It is unknown. An observed landscape $\widehat{L}_t$ estimates this function using past evidence and explicit stability assumptions. No identity makes old average risk equal to future conditional risk under arbitrary drift.

Definition 1.1 (Separate coordinate layers)**.** A landscape layer fixes target semantics, horizon, channel metric, system normalization and model convention. Pockets in different layers remain distinct unless a stated transformation makes their risks and coordinates comparable.

If m means sampling stride, increasing it reduces the number of observations in a window. If it means polynomial degree, increasing it changes model complexity. The same symbol cannot silently switch meanings inside a calculation. Fractional W, m require actual sampling or an interpolation model; interpolation does not add independent observations.

2 Constructing a causal empirical landscape

For point targets, define matured origins $\mathcal{A}_t(P) = \{s : s + P \le t\}$. With nonnegative $\mathcal{F}_t$-measurable comparison weights $w_{t,s}$, set

$$\widehat{L}_t(r) = \frac{\sum_{s \in \mathcal{A}_t(P)} w_{t,s} \ell_T(Y_{s,P}^T, \widehat{Y}_{s,P,r})}{\sum_{s \in \mathcal{A}_t(P)} w_{t,s}}. \tag{2.1}$$

The denominator must be positive. Empty history is missing risk, not zero risk. Every $\widehat{Y}_{s,P,r}$ is the forecast issued using the prefix at s. Refitting a model at t and evaluating its fit on old targets estimates a different quantity.

For a cumulative event, the label may mature before $s + P$ when the event occurs, but a negative label generally requires follow-up through $s + P$. A simple uniform rule retains only complete windows; a survival model can instead handle censoring explicitly. Never label incomplete negative windows healthy merely because no event has yet been seen.

Proposition 2.1 (Information-cutoff preservation)**.** *If all issued forecasts and features at s are $\mathcal{F}_s$-measurable, the stored outcomes in (2.1) are known by t, and the weights are $\mathcal{F}_t$-measurable, then $\widehat{L}_t$ is $\mathcal{F}_t$-measurable. Consequently any deterministic measurable pocket extraction from it is also causal at t.*

Proof. For $s \le t$, $\mathcal{F}_s \subseteq \mathcal{F}_t$. Every summand, its weight and the positive denominator are measurable with respect to $\mathcal{F}_t$. Finite sums, division by a nonzero measurable quantity and measurable composition preserve measurability. □

This proves absence of future information in the construction, not accuracy of the risk estimate. Weights chosen to make a preferred cell look good can still overfit known outcomes.

Worked example 2.2 (A time ledger)**.** At $t = 12$ with $P = 3$, origins $s = 4, 7, 9$ have targets at $7, 10, 12$ and are mature. Origin $s = 10$ has a target at 13 and is excluded. If two lenses have matured losses $(1, 4, 1)$ and $(2, 1, 2)$ with equal weights, their risks are 2 and $5/3$. An unresolved fourth outcome cannot be entered as zero to improve either risk.

3 Noise, model mismatch and signed probes

For a linear historical forecast, let $X = f + \epsilon$, $\mathbb{E}\epsilon = 0$, $\mathrm{Cov}(\epsilon) = \Sigma$. The frozen row q_r defines target minus forecast error. Then

$$\mathbb{E}[(q_r^\top X)^2] = (q_r^\top f)^2 + q_r^\top \Sigma q_r. \tag{3.1}$$

The proof expands the square and uses zero-mean noise. If Q_r stacks rows, the average energy is

$$\mathbb{E}U_r = \frac{\|Q_r f\|^2 + \mathrm{tr}(Q_r \Sigma Q_r^\top)}{q}. \tag{3.2}$$

This explains why pockets depend jointly on the signal, predictor, sampling, noise and horizon. They are not functions of a source equation alone. If the desired target is latent $f(t+P)$ rather than noisy $X(t+P)$, use the corresponding error row and covariance; including future observation noise changes the loss target.

The signed packet $S_r = Q_r X$ retains direction while $U_r = q^{-1}\|S_r\|^2$ does not. $Q_r h = 0$ implies X and $X + h$ have the same packet. Energy also identifies S and $-S$. A landscape made entirely of second differences cannot reveal an affine level or trend without additional anchors.

Application rule. Store the risk definition, probe operator, signs, units and noise assumptions with each layer. A geometric basin is evidence about that forecasting construction, not an automatic physical decomposition.

4 Pocket extraction and a complete geometric record

Let $\mathcal{R}$ be a finite grid with declared adjacency or a continuous domain. At threshold λ, define

$$\mathcal{B}_t(\lambda) = \mathrm{CC}\{r : \widehat{L}_t(r) \le \lambda\}. \tag{4.1}$$

Connected components of a sublevel set are one convention. Attraction basins of minima under a specified descent rule are another; their boundaries need not agree. State the convention and treatment of plateaus and ties.

For a basin B_i, retain its minimum and minimum position, support mask, risk-relative depth, prominence, width, curvature where differentiable, complete sampled flow slices, existence flag and uncertainty. A centroid can be

$$\bar{r}_i = \frac{\sum_{r \in B_i} a_r \omega_r r}{\sum_{r \in B_i} a_r \omega_r}, \tag{4.2}$$

where a_r is cell volume and $\omega_r \ge 0$ is a declared geometric weight. An unweighted centroid and a depth-weighted centroid answer different questions. A nonconvex basin's centroid may lie outside its support.

Worked example 4.1 (Threshold-dependent existence)**.** On a line of five adjacent cells let $L = (3, 1, 4, 2, 3)$. At $\lambda = 2$, the components are cells $\{2\}$ and $\{4\}$. At $\lambda = 3$, they become $\{1, 2\}$ and $\{4, 5\}$. At $\lambda = 4$, they merge. "Two pockets exist" is therefore conditional on a segmentation scale, not an absolute count of source families.

Use a range of thresholds to inspect structural persistence. This is a mathematical organization of basin summaries; it does not turn persistence into a correctness probability.

5 Coordinate transformations and meaningful distances

Let $u = N(r)$ be a one-to-one coordinate map and $\widetilde{L}(u) = L(N^{-1}u)$. Ordering of corresponding risks is unchanged. With a homeomorphism, connected components correspond; numerical width, curvature and Euclidean distances usually change.

For $r = J^{-1}u$ with constant invertible J, gradients and Hessians obey

$$\nabla_u \widetilde{L} = J^{-\top} \nabla_r L, \qquad H_u = J^{-\top} H_r J^{-1}. \tag{5.1}$$

For nonlinear N^{-1}, the Hessian also includes second-derivative terms weighted by $\nabla_r L$; these vanish at an exact stationary point. Thus comparing curvature across coordinate conventions without transformation is invalid.

A metric $d(r, r')^2 = (r - r')^\top M(r - r')$ can encode units and relative scales. Under $u = Jr$, use $M_u = J^{-\top} M J^{-1}$ to preserve distance. Standardizing using the entire future dataset is not causal; fit or specify normalizers before issuance.

Worked example 5.1 (Units change nearest-neighbor assignments)**.** Suppose one coordinate is measured in seconds and another in milliseconds. Raw Euclidean distance gives the millisecond coordinate a thousandfold numerical scale. Converting units or using a corresponding metric repairs that artifact. It does not prove that the two forecasting targets have the same meaning.

6 Stability bounds for landscape use

Theorem 6.1 (Sublevel inclusions)**.** *If* $\left\|\widehat{L} - L\right\|_\infty \le \varepsilon$*, then*

$$\{L \le \lambda - \varepsilon\} \subseteq \{\widehat{L} \le \lambda\} \subseteq \{L \le \lambda + \varepsilon\}. \tag{6.1}$$

Proof. For the left inclusion, $L(r) \le \lambda - \varepsilon$ implies $\widehat{L}(r) \le L(r) + \varepsilon \le \lambda$. For the right inclusion, $\widehat{L}(r) \le \lambda$ implies $L(r) \le \widehat{L}(r) + \varepsilon \le \lambda + \varepsilon$. □

These inclusions bound threshold uncertainty, but do not establish a one-to-one matching of components. A shallow bridge can appear or disappear inside the uncertainty band.

Theorem 6.2 (Selection error and an isolated winner)**.** *Let* r_* *minimize* L *on a finite candidate set, and let* $\widehat{r}$ *minimize* $\widehat{L}$*. Under the preceding uniform bound,*

$$L(\widehat{r}) - L(r_*) \le 2\varepsilon. \tag{6.2}$$

If the true gap from r_* *to every other cell exceeds* 2ε*, then* $\widehat{r} = r_*$*.*

Proof. $L(\widehat{r}) \le \widehat{L}(\widehat{r}) + \varepsilon \le \widehat{L}(r_*) + \varepsilon \le L(r_*) + 2\varepsilon$. If another cell were selected despite a larger gap, the same inequality would be contradicted. □

When the gap is small, report a set of near-optimal cells rather than a falsely precise optimum. Preserving several basins can represent resolution uncertainty; it does not prove that mixing their forecasts is better.

Proposition 6.3 (A grid-resolution bound)**.** *Suppose* L *is* K*-Lipschitz on a compact domain and a grid lies within distance* h *of every point. Then* $0 \le \min_{r \in \mathcal{R}_h} L(r) - \min_r L(r) \le Kh$*.*

Proof. Choose a grid point within h of a continuous minimizer and apply Lipschitz continuity. The grid minimum cannot be lower than the unrestricted minimum. □

This controls minimum value, not basin count or the best cell's location. A finer grid can create more numerically distinguishable minima without adding measurements.

7 Statistical uncertainty of an empirical surface

Let N fixed candidate predictors be evaluated on n independent validation examples, with losses in $[0, 1]$. Hoeffding's inequality and the union bound give, with probability at least $1 - \delta$,

$$\max_r |\widehat{L}(r) - L(r)| \le \sqrt{\frac{\log(2N/\delta)}{2n}}. \tag{7.1}$$

The proof applies the two-sided bound $2e^{-2n\varepsilon^2}$ to each candidate and sums over N. Conditional on independent training data, the result also holds for predictors fixed by that training data. It does not hold unchanged when predictors or the candidate set are repeatedly selected on the same validation sample, when losses are unbounded, or when overlapping temporal outcomes are treated as independent.

For independent homoscedastic losses and deterministic weights, the weighted mean has variance $\sigma_\ell^2 \sum_s w_s^2/(\sum_s w_s)^2$. This motivates $n_{\text{eff}} = (\sum_s w_s)^2/\sum_s w_s^2$, but correlation adds terms:

$$\operatorname{Var}\left(\sum_s \alpha_s \ell_s\right) = \sum_{s,u} \alpha_s \alpha_u \operatorname{Cov}(\ell_s, \ell_u). \tag{7.2}$$

The independence-based effective size is not a general correction for time dependence. Use independent runs as units when appropriate, or explicitly justified temporal dependence assumptions. Risk estimation must match the intended future population.

8 An equation trail for window, curvature and noise

Consider the local signal $f(u) = \beta_0 + \beta_1 u + \kappa u^2/2$ on historical $u \in [-W, 0]$. Fit a line by uniform continuous least squares. The projection of u^2 onto $\{1, u\}$ is $-W^2/6 - Wu$: its residual is orthogonal to both 1 and u, as direct integration verifies. The fitted line is therefore

$$\widehat{f}(u) = \beta_0 - \kappa W^2/12 + (\beta_1 - \kappa W/2)u. \tag{8.1}$$

At future $P \ge 0$ the approximation bias is

$$f(P) - \widehat{f}(P) = \kappa\left(P^2/2 + WP/2 + W^2/12\right). \tag{8.2}$$

With n roughly uniform observations, independent noise variance σ^2 and the continuous-design approximation, latent-target prediction variance is

$$v(W, P) \simeq \frac{\sigma^2}{n}\left[1 + \frac{12(P + W/2)^2}{W^2}\right]. \tag{8.3}$$

The risk approximation is bias squared plus v. To derive the variance, center the linear regressor at $-W/2$ and use its variance $W^2/12$. The intercept and centered slope are orthogonal in this approximation.

At $P = 0$, with spacing Δ and $n \simeq W/\Delta$, this becomes $R(W) \simeq \kappa^2 W^4/144 + 4\sigma^2\Delta/W$. Differentiation gives

$$W_*^5 = 144\sigma^2\Delta/\kappa^2 \qquad (\kappa \neq 0). \tag{8.4}$$

This restricted boundary-estimation calculation explains a fifth-root trade-off. It is not a universal WPM equation. At $P > 0$, correlated noise, different model degree, regime changes or a different interpretation of m, the objective changes. If $\kappa = 0$, there is no finite optimum from this formula alone.

9 Temporal history, identity and optional dynamics

At each time retain individual basin records and their support, flow curves and existence. A lineage graph has directed links between plausible historical parent and child pockets. One possible matching cost is

$$c_{ij} = a\, d(z_i, z_j) + b\,[1 - \mathrm{IoU}(B_i, B_j)] + c\, d_{\mathrm{flow}}(i, j). \tag{9.1}$$

Coefficients and scales are modeling choices. Allow uncertain birth, disappearance, split and merge. A switch of the global winning cell between persistent basins is a selection change, not movement of one basin. Averaging two identities can manufacture a track that belongs to neither.

If one proposes a pocket-state law $\dot{z} = v_\theta(z, C)$, a numerical integrator only approximates that supplied law. It does not infer v_θ from nothing. Existence should be modeled separately, for example $\mathbb{P}(e_{t+h} = 1 \mid H_t)$, before scoring a position conditional on survival. A very accurate conditional position forecast may cover few pockets if most disappear.

10 Forecasting a field and interpreting probability maps

An observed sequence of carrier fields can support a model $V_{t+P} = \mathcal{G}_\theta(V_{\le t}, C_t, P) + \epsilon$. This is a new prediction problem. Knowing an exact current composition operator does not determine future inputs to that operator.

For an event $E(x, P)$, let $p_t(x, P) = \mathbb{P}(E(x, P) \mid \mathcal{F}_t)$. A local occurrence-probability map need not integrate to one. A density for one random location does integrate to one with respect to its declared measure. For events on a finite grid,

$$\max_x p_t(x, P) \le \mathbb{P}\left(\bigcup_x E(x, P) \mid \mathcal{F}_t\right) \le \min\left(1, \sum_x p_t(x, P)\right). \tag{10.1}$$

Independence would permit a product formula, but correlated spatial or temporal events require their joint law. Probability-map scoring should use a proper score and the event definition [4]; a colorful map alone has no calibration guarantee.

11 A landscape worksheet from errors to decisions

Take three windows $W \in \{2, 4, 6\}$ and strides $m \in \{1, 2, 3\}$ at one fixed target and horizon. Suppose matured issued errors produce this empirical squared-risk table:

	$m = 1$	$m = 2$	$m = 3$
$W = 2$	0.90	0.30	0.80
$W = 4$	0.70	0.60	0.90
$W = 6$	0.40	0.80	1.00

For the first minimum, three errors $(-0.5, 0.5, \sqrt{0.4})$ yield mean squared risk $(0.25{+}0.25{+}0.4)/3 = 0.30$. For the second, errors $(0.6, -0.6, \sqrt{0.48})$ yield 0.40. The signs are retained separately from the risk table. They can affect error-covariance combination even though segmentation uses nonnegative risks.

Using four-neighbor adjacency and threshold 0.45, two pockets exist: single cells $(2, 2)$ and $(6, 1)$. The complete W-flow through the first is $(0.30, 0.60, 0.80)$ at $m = 2$; its m-flow is $(0.90, 0.30, 0.80)$ at $W = 2$. The first is a short-window preference, not a source moving through physical space. The second pocket's W-flow is $(0.90, 0.70, 0.40)$ at $m = 1$. Retaining the complete slices exposes these different shapes.

If a uniform risk-error bound were 0.08, the observed winning gap 0.10 would be below $2\varepsilon = 0.16$. Indeed, a candidate at true risk 0.38 could be observed as 0.30 while another at 0.32 could be observed as 0.40. Thus the winner is unresolved under that error budget. A set of near-optimal candidates is a more defensible representation than reporting one exact resolution as known.

Suppose next that the two pocket-associated current forecasts are 10 and 14. Their risk heights do not determine a unique forecast mixture. Weights proportional to inverse risk would be a policy choice, not a theorem. A second-moment matrix of their matured forecast errors, a distributional model or an explicitly learned router is needed to justify a combination. Nor does a current minimum at $(6, 1)$ predict the next minimum's location without a temporal model.

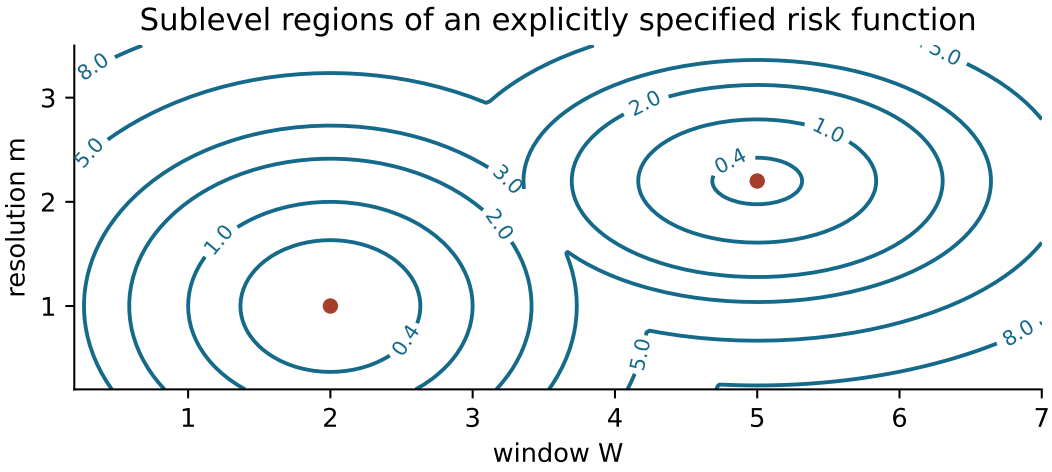


Figure 1: An analytical two-basin landscape, $L(W, m) = \min\{(W-2)^2 + (m-1)^2,\ 0.3 + (W-5)^2 + 2(m-2.2)^2\}$. Contours are risk thresholds, and the marked minima are distinct pockets at sufficiently low thresholds. This function is a constructed geometry example, not a fitted risk surface.

12 Independent layers, missing data and topology

A grid is a sampling of an objective, not an exchangeable dataset of independent risks. Adjacent cells often reuse the same history and target outcomes. Do not bootstrap cells to obtain a sample size equal to the number of W,m settings. If uncertainty is assessed across independent runs, each resampled unit must retain its full table and dependent outcomes.

Missing cells require a mask. Connecting two observed low-risk cells through an unobserved region assumes something about that region. A conservative extractor can refuse connectivity across missing cells; an interpolating extractor must record an interpolation model and its uncertainty.

Plotting software that fills missing cells for a smooth image has not measured their risks.

Independent horizon or model-degree layers should not be merged just because their minima have similar coordinates. Loss magnitudes may differ by horizon, target dimension or normalization. If cross-layer comparison is desired, define a common decision objective or a lawful transformation first. A monotone transform of L preserves ordering within a layer but changes arithmetic combinations and usually changes depth and prominence values.

There is also a distinction between relative and absolute depth. If every candidate becomes worse by the same constant, relative pocket geometry can remain unchanged while forecast quality deteriorates. Retain absolute minimum risk and a reference predictor alongside relative basin shape. Conversely, a global rescaling of measurement units can multiply all squared losses without changing the underlying predictive quality after decoding.

13 From geometric stability to a responsible report

The uniform-error bounds in this paper separate three questions: is the selected risk value close to optimal, is the winning cell identified, and are the extracted pockets topologically stable? The first has a 2ε bound; the second requires a gap; the third requires additional separation or topological assumptions. One should not replace these distinctions with a single "landscape accuracy" score.

A reproducible mathematical specification therefore includes the forecast query, issued-prediction ledger, candidate domain, risk weights, missing mask, coordinate metric, adjacency and threshold rule. A temporal specification also includes matching costs, birth/death handling and uncertainty of identity. If a future landscape is predicted, identify the model and whether its output is a point surface, a distribution over surfaces or a set with a stated interpretation.

14 Application protocol and exercises

1. Define target, horizon, units, information set and loss before drawing a landscape.
2. Enumerate lenses with explicit W, m meanings. Reconstruct historical forecasts at their actual issuance cutoffs.
3. Compute matured risks and mark missing cells. Store uncertainty and dependence information.
4. Declare adjacency, segmentation and coordinate metric. Retain full flow slices, locations and existence, rather than only the deepest cell.
5. Check sensitivity to grid, threshold and risk error. Keep near-optimal alternatives when the gap is unresolved.
6. Track plausible identities over time; attach a separate predictive model if a future landscape or target is required.

Exercise 14.1. A true three-cell risk is $(0.20, 0.23, 0.40)$ and uniform error is at most 0.02. Is the best cell certified?

Solution. No: the gap 0.03 is below $2\varepsilon = 0.04$. The first two estimated values can reverse. The selection-risk penalty is nevertheless bounded by 0.04 under the uniform assumption.

Exercise 14.2. If L is 3-Lipschitz and the grid covering radius is 0.01, what is the minimum-value discretization error bound?

Solution. At most 0.03. This says nothing about uniqueness of the minimizer.

Exercise 14.3. Two future-time occurrence events are identical and each has probability 1/2. Compute their union probability and compare it with the independence shortcut.

Solution. The union equals the same event and has probability 1/2. The shortcut gives $1 - (1/2)^2 = 3/4$, demonstrating why marginal maps cannot supply a general joint probability.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This paper and its companion (Papers 1 and 2 of the series) share the archival DOI 10.5281/zenodo.23781662.

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